

Anantram Patel



Associate Data Scientist at 3D Engineering Automation | working Location: ADE, DRDO, Bengaluru
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Summary

Proficient in linear algebra and statistical analysis. Data Scientist familiar with gathering, cleaning and organizing data for use by technical and non-technical personnel. Advanced understanding of statistical, algebraic and other analytical techniques. Motivated and diligent with significant background in Deep Neural Network (DNN).

Technical Skill

Python, Java, SQL, Spring Boot, PyTorch, Tensorflow, Keras, NLTK, Scikit-Learn, NumPy, SciPy, PyQt5, NetworkX, Pandas, Matplotlib, Seaborn, Manim, Jupyter Notebook, VS Code, PyCharm.

Competencies

Linear Algebra, Statistics, Data Visualization, Feature Engineering, Machine Learning (ML), Deep Learning (DL), Natural language processing(NLP), Data Engineering, Data Modelling, Transfer Learning, Graphical User Interface, Software Development.

Work History

Associate Data Scientist | 06-2022 – Current

3D Engineering Automation llp, Bangalore

- Visualization and analytics to analyze and process complex data by the help of the help of python.
- Collaborated on the development of big data solution using Python, NumPy improving data processing.
- Leading the team data professionals to drive data-based decision-making efforts.
- Compiled, cleaned and manipulated data for proper handling, Modeled predictions with feature selection.

Research Engineer | 01-2022 – 06-2022

Bangalore Aircraft Industries Pvt. Ltd., Bangalore

- Performed training and testing of engineering models to support adequacy and reformulated models.
- Extensively used python library like NumPy, pandas, matplotlib, seaborn for data visualization.
- Conducted engineering and detailed experimental tests to collect design data and assist in research work.

Freelancer Mathematics and Python | 03-2019 – 10-2021

Chegg, Bhopal

- Calculus, Linear Algebra, Statistic, Engineering Mathematics, problem-solving, critical thinking
- Self-motivated, with strong sense of personal responsibility

Research Papers

Deep Learning Based Graph Neural Network Technique for Hardware Trojan Detection at Register Transfer Level

A novel golden reference-free HT detection method for both Register Transfer Level (RTL) and gate-level netlists by leveraging Graph Neural Networks (GNNs) to learn the behavior of the circuit through a Data Flow Graph (DFG) representation of the hardware design.

Projects

Text clustered GNN: Fake news detection on social media.

Developed a novel approach for fake news detection that constructs a detection framework solely from text, avoiding reliance on auxiliary data. Utilized text clustering to extract representative words and employed multiple clustering dimensions to enhance semantic representation. Model accuracy is increased by 6% from the existing other models.

Artificial neural network based inverse design: Airfoils and wings

In this project, we compare Convolutional Neural Network (CNN) and Deep Neural Networks (DNN) to inverse design the airfoil shapes. The pressure distribution over the airfoil in pictorial (for CNN) or numerical form (for DNN), the trained neural networks predict the airfoil shapes. the DNN model is found to be $\approx 7\times$ faster than the CNN. The accuracy of DNN is also observed to be superior to that of CNN.

Artificial Intelligence Based High-Frequency Electromagnetic Simulation of Aircraft

In this research, I developed some machine learning techniques to predict RCS value at the given angle and frequency on the aircraft. I developed one error correction techniques which is applied on predicted RCS value.

Tweets multiclass classification based on sentiment analysis

Classifying tweets based on positive, negative and neutral sentiment. Data cleaning, missing value handling, feature selection, feature engineering etc.

Education

Bachelor of Engineering: Electrical Engineering | CGPA: 6.57 | 2014 – 2018

Government Engineering Collage Rewa - Madhya Pradesh, India

12th MP Board | Mark: 82% | 2012-2013

Government Higher Secondary School Ghooma - Madhya Pradesh, India

10th MP Board | Mark: 80% | 2011-2012

Government Higher Secondary School Ghooma - Madhya Pradesh, India